

NICHOLAS RUFFIN

Principal Data Engineer | Data Architecture & Strategy

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PROFESSIONAL SUMMARY

Data engineer with 12+ years of experience defining data architecture, setting long-term technical strategy, and building the platforms that drive business outcomes. Currently own the end-to-end architecture of a data monetization platform generating \$7M+ ARR, processing 4.8B de-identified records through Snowflake and AWS. Prior experience as a Director of Engineering managing 16+ people provides a unique ability to partner with executive leadership, align technical decisions to business goals, and influence across teams without direct authority. Deep hands-on expertise in Python, SQL, AWS, Airflow, and cloud data warehouse optimization.

TECHNICAL SKILLS

Data Architecture	Data Warehouse Design, ELT/ETL Pipeline Architecture, Data Modeling, Data Quality & Observability, Platform Scalability
Engineering	Python, SQL, Snowflake (Procs, Streams, Tasks), AWS (MWAA, S3, Lambda, IAM), Kafka, AWS Glue
Orchestration	Apache Airflow, Terraform, GitHub Actions (CI/CD), Docker, Git
Strategy	Technical strategy, cross-team influence, vendor evaluation, TCO analysis, HIPAA compliance, Datavant tokenization

EXPERIENCE

Senior Clinical Data Engineer RedSail Technologies	Feb 2025 - Present
- Own the end-to-end architecture of a data monetization platform generating \$7M+ ARR, designing secure PII tokenization workflows (Snowflake + Datavant) that enable pharmaceutical clients to access privacy-compliant clinical datasets. - Defined the platform's data architecture and long-term technical strategy, establishing Snowflake as the single source of truth processing 4.8 billion records (3M+ transactions/hour) with strict HIPAA compliance. - Built Python/Airflow orchestration framework deployed via GitHub Actions, automating 50+ daily workflows for S3 extraction and ELT processing. - Optimized cloud data warehouse architecture reducing query latency by 40%, implemented data quality monitoring achieving 99.5% pipeline uptime SLA. - Partner with product and business stakeholders to align platform roadmap with revenue goals, translating business requirements into scalable data solutions.	
Director of Full Stack Development Nimble Solutions	Aug 2021 - Jan 2025

Promoted from Lead Software Engineer to Director by the CIO. Built and managed 16+ engineers across three teams: 5 US software engineers, 6 SQL/data developers, 5 offshore developers in Hyderabad, and a project manager.

- Defined technical strategy and architecture across all engineering teams, partnering with CIO, VP of Infrastructure, and CISO on **vendor evaluations, TCO analysis**, and long-term platform direction.
- Led technology due diligence for **4 acquisitions**, evaluating data systems, technical debt, and integration paths. Traveled to Hyderabad to integrate the acquired development team into US operations.
- Architected **QUARC**, a queue-based AR claims reimbursement system with historical AR aging data, increasing throughput to **30+ claims/day per rep** — designed the data model, pipeline architecture, and workflow orchestration.
- Engineered integration of **50+ data sources** into a unified enterprise data architecture, designing API orchestration patterns for high reliability and scalability.
- Established engineering standards, implemented CI/CD pipelines reducing deployment time by **60%**, and drove **90%+ test code coverage** across all teams.

Senior Full Stack Developer | Core & Main

Jun 2017 - Aug 2021

- Developed enterprise data applications using C#/.NET and SQL, supporting **200+ concurrent** corporate users.

EDUCATION

Bachelor of Science, Information Technology

University of Phoenix